torch.linalg.inv#

https://pytorch.org/docs/stable/generated/torch.linalg.inv.html

Description:

Computes the inverse of a square matrix if it exists. Throws a RuntimeError if the matrix is not invertible. Letting K\mathbb{K}K be R\mathbb{R}R or C\mathbb{C}C, for a matrix A∈Kn×nA \in \mathbb{K}^{n \times n}A∈Kn×n, its inverse matrix A−1∈Kn×nA^{-1} \in \mathbb{K}^{n \times n}A−1∈Kn×n (if it exists) is defined as where In\mathrm{I}_nIn is the n-dimensional identity matrix. The inverse matrix exists if and only if AAA is invertible. In this case, the inverse is unique. Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions.

Function Signature

```
torch.linalg.inv(A, *, out=None) → Tensor#

Parameters

Keyword Arguments

Raises
```

Parameters

Keyword

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

Code Examples

```
linalg.solve(A, B) == linalg.inv(A) @ B  # When B is a matrix
```

```
>>> A = torch.randn(4, 4)
>>> Ainv = torch.linalg.inv(A)
>>> torch.dist(A @ Ainv, torch.eye(4))
tensor(1.1921e-07)

>>> A = torch.randn(2, 3, 4, 4)  # Batch of matrices
>>> Ainv = torch.linalg.inv(A)
>>> torch.dist(A @ Ainv, torch.eye(4))
tensor(1.9073e-06)

>>> A = torch.randn(4, 4, dtype=torch.complex128)  # Complex
matrix
>>> Ainv = torch.linalg.inv(A)
>>> torch.dist(A @ Ainv, torch.eye(4))
tensor(7.5107e-16, dtype=torch.float64)
```

Notes & Warnings

NOTE Note When inputs are on a CUDA device, this function synchronizes that device with the CPU. For a version of this function that does not synchronize, see torch.linalg.inv_ex().

NOTE Note Consider using torch.linalg.solve() if possible for multiplying a matrix on the left by the inverse, as: linalg.solve(A, B) == linalg.inv(A) @ B # When B is a matrix It is always preferred to use solve() when possible, as it is faster and more numerically stable than computing the inverse explicitly.

NOTE See also torch.linalg.pinv() computes the pseudoinverse (Moore-Penrose inverse) of matrices of any shape. torch.linalg.solve() computes A.inv() @ B with a numerically stable algorithm.

Detailed Documentation

Documentation

Computes the inverse of a square matrix if it exists. Throws a RuntimeError if the matrix is not invertible.

Letting K\mathbb{K}K be R\mathbb{R}R or C\mathbb{C}C, for a matrix A∈Kn×nA \in

\mathbb{K}^{n \times n}A∈Kn×n, its inverse matrix A−1∈Kn×nA^{-1} \in \mathbb{K}^{n \times n}A−1∈Kn×n (if it exists) is defined as

where In\mathrm{I}_nIn is the n-dimensional identity matrix.

The inverse matrix exists if and only if AAA is invertible. In this case, the inverse is unique.

Supports input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices, and if A is a batch of matrices then the output has the same batch dimensions.

When inputs are on a CUDA device, this function synchronizes that device with the CPU. For a version of this function that does not synchronize, see torch.linalg.inv_ex().

Consider using torch.linalg.solve() if possible for multiplying a matrix on the left by the inverse, as:

It is always preferred to use solve() when possible, as it is faster and more numerically stable than computing the inverse explicitly.

torch.linalg.pinv() computes the pseudoinverse (Moore-Penrose inverse) of matrices of any shape.

torch.linalg.solve() computes A.inv() @ B with a numerically stable algorithm.

A (Tensor) – tensor of shape (*, n, n) where * is zero or more batch dimensions consisting of invertible matrices.

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

RuntimeError – if the matrix A or any matrix in the batch of matrices A is not invertible.